

Introduction to information & communication technologies (IICT)

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Why this Subject?

- Introductory course
- Guide those who have less/no familiarity with computer
- Know the past, present & future of Computer.

What in this subject? (Course Outline)

Brief history of Computer, Four Stages of History, Computer Elements, Processor, Memory, Hardware, Software, Application Software its uses and Limitations, System Software its Importance and its Types, Types of Computer (Super, Mainframe, Mini and MicroComputer), Introduction to CBIS (Computer Based Information System), Methods of Input and Processing, Class2. Organizing Computer Facility, Centralized Computing Facility, Distributed Computing Facility, Decentralized Computing Facility, Input Devices. Keyboard and its Types, Terminal (Dumb, Smart, Intelligent), Dedicated Data Entry, SDA (Source Data Automation), Pointing Devices, Voice Input, Output Devices. Soft- Hard Copies, Monitors and its Types, Printers and its Types, Plotters, Computer Virus and its Forms, Storage Units, Primary and Secondary Memories, RAM and its Types, Cache, Hard Disks, Working of Hard Disk, Diskettes, RAID, Optical Disk Storages (DVD, CD ROM), Magnetic Types, Backup System, Data Communications, Data Communication Model, Data Transmission, Digital and Analog Transmission, Modems, Asynchronous and Synchronous Transmission, Simplex, Half Duplex, Full Duplex Transmission, Communications, Medias (Cables, Wireless), Protocols, Network Topologies (Star, Bus, Ring), LAN, WAN, Internet, A Brief History, Birthplace of ARPA Net, Web Link, Browser, Internet Services provider and Online Services Providers, Function and Features of Browser, Search Engines, Some Common Services available on Internet.

Reference Material

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2. Livesley, Robert Kenneth. An introduction to automatic digital computers. Cambridge University Press, 2017.
3. Zawacki-Richter, Olaf, and Colin Latchem. "Exploring four decades of research in Computers & Education." Computers & Education 122 (2018): 136-152.
4. Sinha, Pradeep K., and Priti Sinha. Computer fundamentals. BPB publications, 2010.
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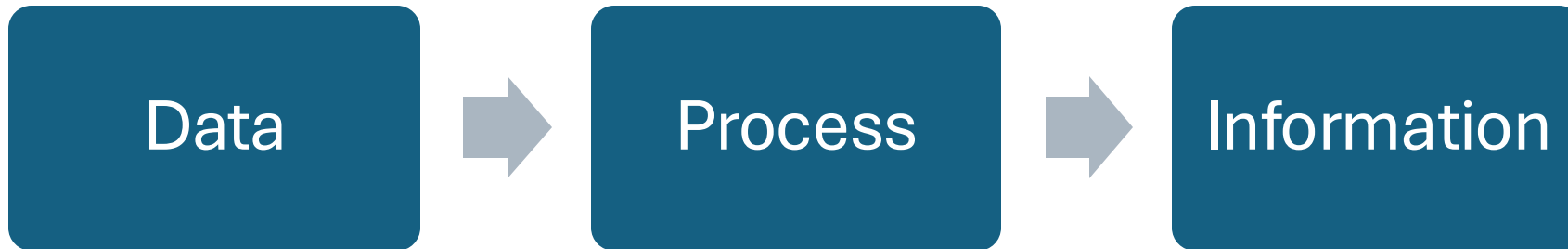


BRIEF HISTORY OF COMPUTER

What is computer, 5 generations of
computer

What is Computer?

- "Compute" => Computer
- Device that performs arithmetic operations upon **Data**.
- Numbers, patterns ,letters, symbol, size.
- A computer can store, process and retrieve data as & when we desire.



What is a computer?

- "A Computer is an electronic device that takes input from a user, processes it and give the output according to user's requirement."
- Q: what is the I.Q. of a computer?



Brief History of Computers

- Set of Technologies is called GENERATION.
- Framework for the growth of computer technology.
- 5 Generations till date.



First Generation (1940-1955)

- Technology: Vacuum tube technology
- Employed vacuum tubes for circuitry and magnetic drums for storage
- Purpose: Used as a calculating device.
- Speed: Performed calculations in milliseconds.
- Size: Too bulky in size & complex design.
- Required large room to place it.
- Cons: Generates too much heat & burnt.
- Required continuously hardware maintenance.
- Generates much heat so must air-conditioner rooms are required.
- Commercial production is difficult & costly.
- Difficult to configure.
- Limited commercial use.
- Examples: ENIAC, EDVAC

<https://www.geeksforgeeks.org/first-generation-of-computer/>

Second Generation (1956-1963)

- Technology: Transistors
 - Employed transistors for process and magnetic cores for storage
- Purpose: Used as a calculating device
- Speed: Performed calculations in microseconds.
- Size: 10x smaller than 1st generation.
 - Required smaller room to place it.
- Cons: Generates heat & need cooling system.
 - Required continuously hardware maintenance.
 - Commercial production is difficult & costly.
 - Limited commercial use.
- Examples: BM 1920, IBM 7094, CDC 1604, CDC 3600, IBM 1401

https://www.geeksforgeeks.org/second-generation-of-computers/?ref=oin_asr6

Rewise

- What replaced vacuum tubes?
- Which language is used for programming in second generation computers?
- What are the advantages of transistors over vacuum tubes?

Third Generation (1965-1971)

- Technology: combination of transistors or Integrated Circuits (IC)
- Purpose: Used as a calculating device
- Speed: Performed calculations in nanoseconds.
- Size: 10x smaller than 2nd generation.
 - Required smaller room to place it.
- Cons: Generates heat & need cooling system.
 - Require less hardware maintenance.
 - Widely used for commercial purpose.
- Examples: IBM 370, PDP-11, IBM System/360, Honeywell-6000, DEC series, and ICL 2900.

Fourth Generation (1975-1989)

- Technology: Microprocessor chip, LSI, VLSI
- Purpose: Used as general-purpose computer.
- Speed: Performed calculations in nanoseconds.
- Size: 1000x smaller than 1st generation.
- Pros: Generates minimal heat & need no cooling system.
- Minimum hardware maintenance required.
- Widely used for commercial purpose as were cheap.
- Examples: STAR 1000, CRAY-X-MP(Super Computer), DEC 10, PDP 11, and CRAY-1

Fifth Generation (1989-now)

- Technology: ULSI
- Purpose: Used as general-purpose computer with user-friendly interface and multimedia feature.
- Speed: Fastest among all previous generations.
- Size: 1000x smaller than 1st generation.
- Pros: Minimum hardware maintenance required.
 - Small & Handy.
 - Consumed least power.
 - High level language
 - Widely used for commercial purpose as were cheap.
- Examples: PARAM 10000, IBM notebooks, Intel P4, Laptops.

Comparision

Generation	1st	2nd	3rd	4th	5th
Era	1940-1955	1956-1963	1965-1971	1975-1989	1989-Till date
Technology	Vacuum Tube	Transistor	IC	LSI,VLSI	ULSI
Size	Room	-10x room	-100x room	-100x room	-1000x room
Speed	Milliseconds	Microseconds	Nanoseconds	Faster than old	Fastest among all
Language	Machine language	Assembly language	FORTAN / COBOL	BASIC/ PASCAL/ COBOL/ FORTAN	Multiple languages
Purpose	Basic calculations	Calculating device	Calculating device	General purpose	General purpose with multimedia feature
Example	ENIAC	IBM 1401	IBM 370	STAR 1000	PARAM 10000



FOUR STAGES OF HISTORY OF INFORMATION TECHNOLOGY

The Four ages

- The Pre-mechanical Age: 3000 BC and 1450 AD
- The Mechanical Age: 1450 – 1840
- The Electromechanical Age: 1840 - 1940.
- The Electronic Age: 1940 - Present.

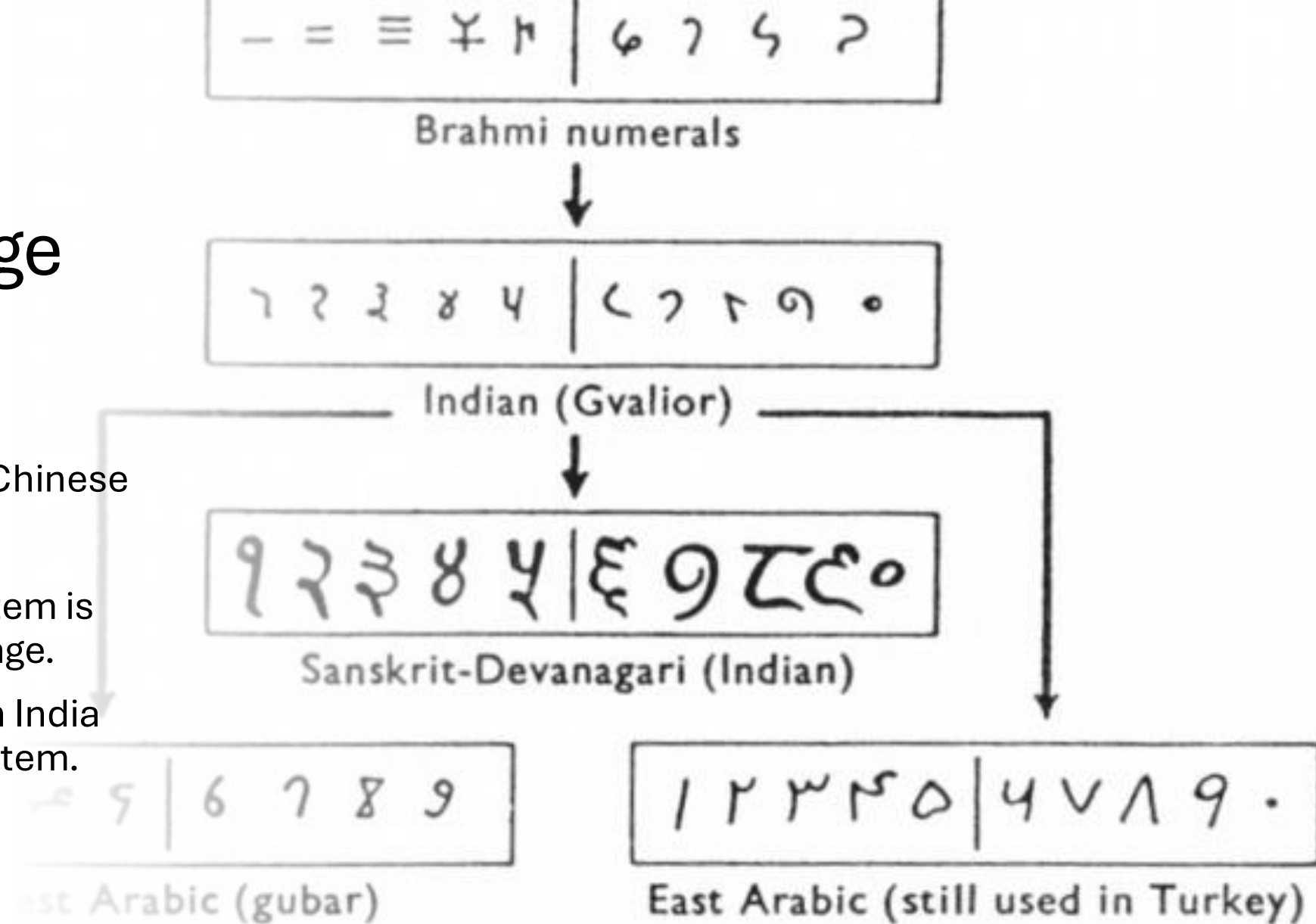
The pre-mechanical age

- It is the earliest age of information technology.
- Between the years 3000 BC and 1450 AD
- Its first form: Writing and Alphabets
- Cuneiform (around 3000 BC), then symbols (around 2000 BC).



The pre-mechanical age

- Papers and Pens
- Sumerians, Egyptians, Chinese
- Books and Libraries
- The first numbering system is also discovered in this age.
- Egyptians and Hindus in India used the numbering system.



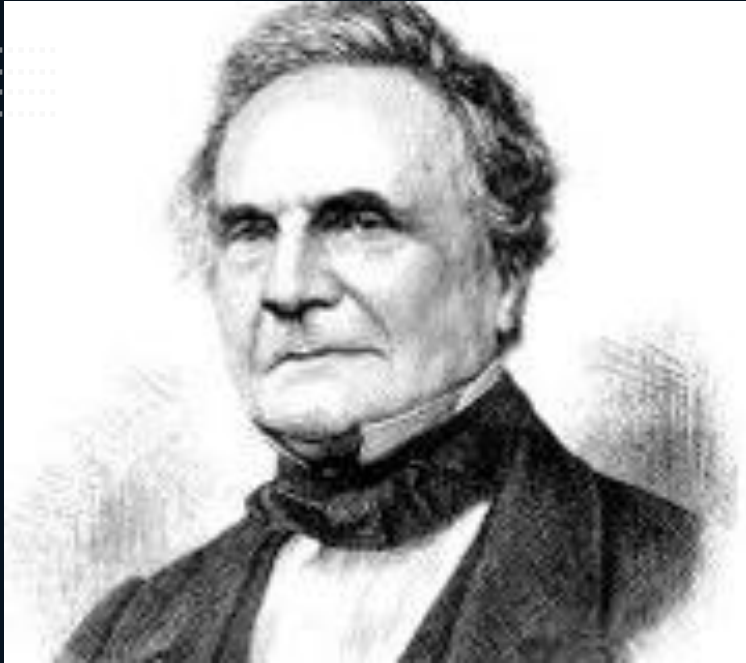


The pre-mechanical age

- The numbering system similar to those we use today is invented between 100 and 200 AD.
- The invention of zero comes much later.
- The first calculator is also discovered during this age.

The Mechanical Age

- This is the age that we observe the first connections between the technology of today and its ancestors.
- Slide rule is invented (early example of analog computers).
- The first general purpose computers are developed.
- "Computer: one who works with numbers."



The Mechanical Age

- Blaise Pascal invented the Pascaline machine.
- Difference engine is invented by Charles Babbage.
- Ada Lovelace is the first programmer.

The electromechanical age

- The most important advance in this era is to use electricity.
- The developments in this age are the beginnings of communication.
- Telegraph (early 1800s), radio
- Morse Code
- Telephone (examples in 1876, 1930, 1970 respectively).



A ●- -	J ●- - -	S ●●●
B - ●●●	K - ●- -	T -
C - ●- -	L - ●●●	U ●●- -
D - ●●●	M - -	V ●●●- -
E ●	N - ●	W ●- - -
F ●●●- ●	O - - -	X - ●●●
G - - ●	P ●- - ●	Y - ●- -
H ●●●●	Q - - ●- -	Z - - ●●
I ●●	R ●- ●	



The Electronic age

- That is the age that we are currently living in.
- ENIAC – the first high-speed, digital computer capable of being reprogrammed to solve a full range of computing problems.
- Used vacuum tubes – no mechanical devices.

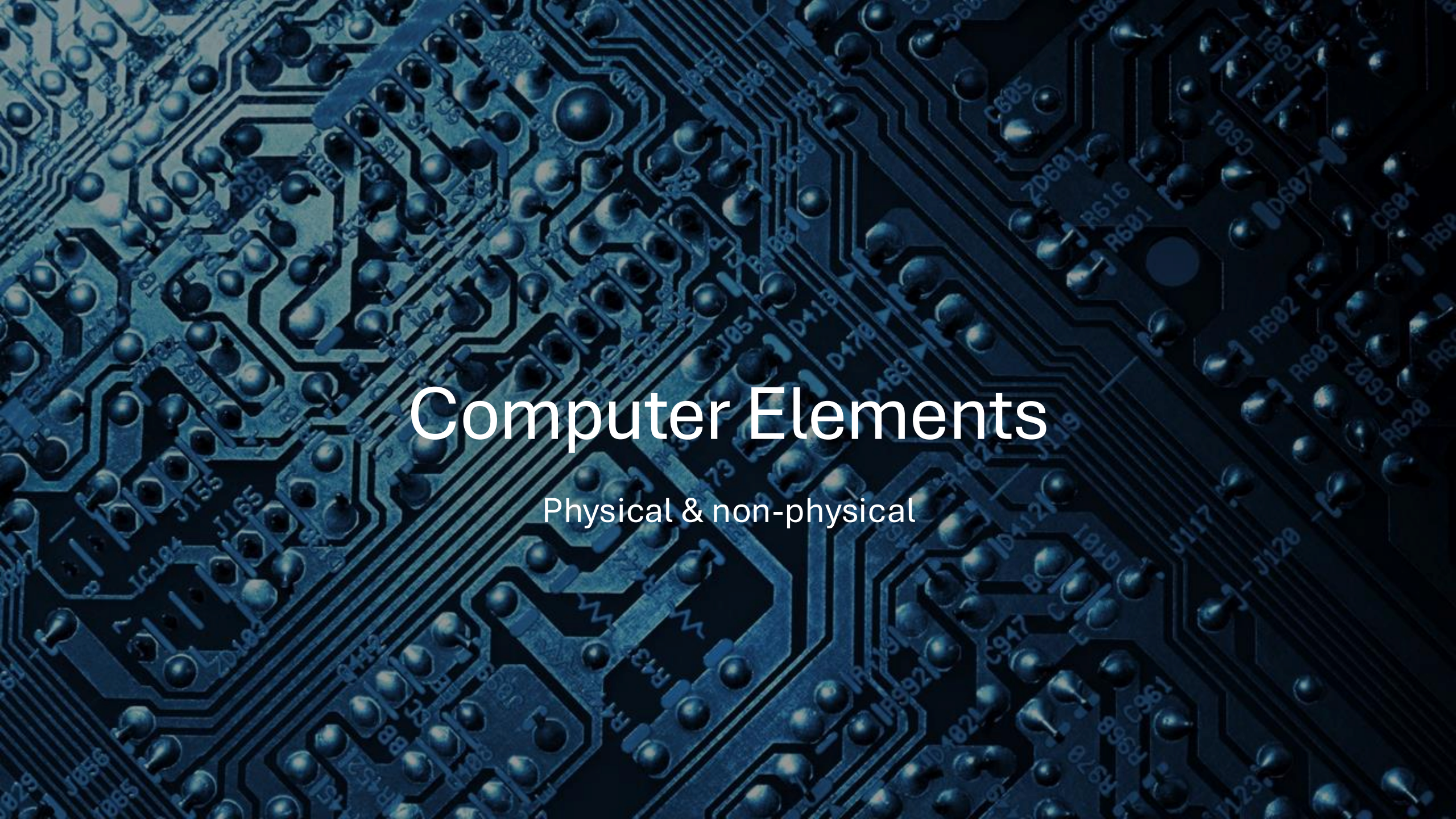
The Electronic Age

- No storage of programs.
- The first stored program computer(s): The University of Manchester Mark I and The EDSAC in Cambridge in 1949.
- The First General-Purpose Computer for Commercial Use: Universal Automatic Computer (UNIVAC)

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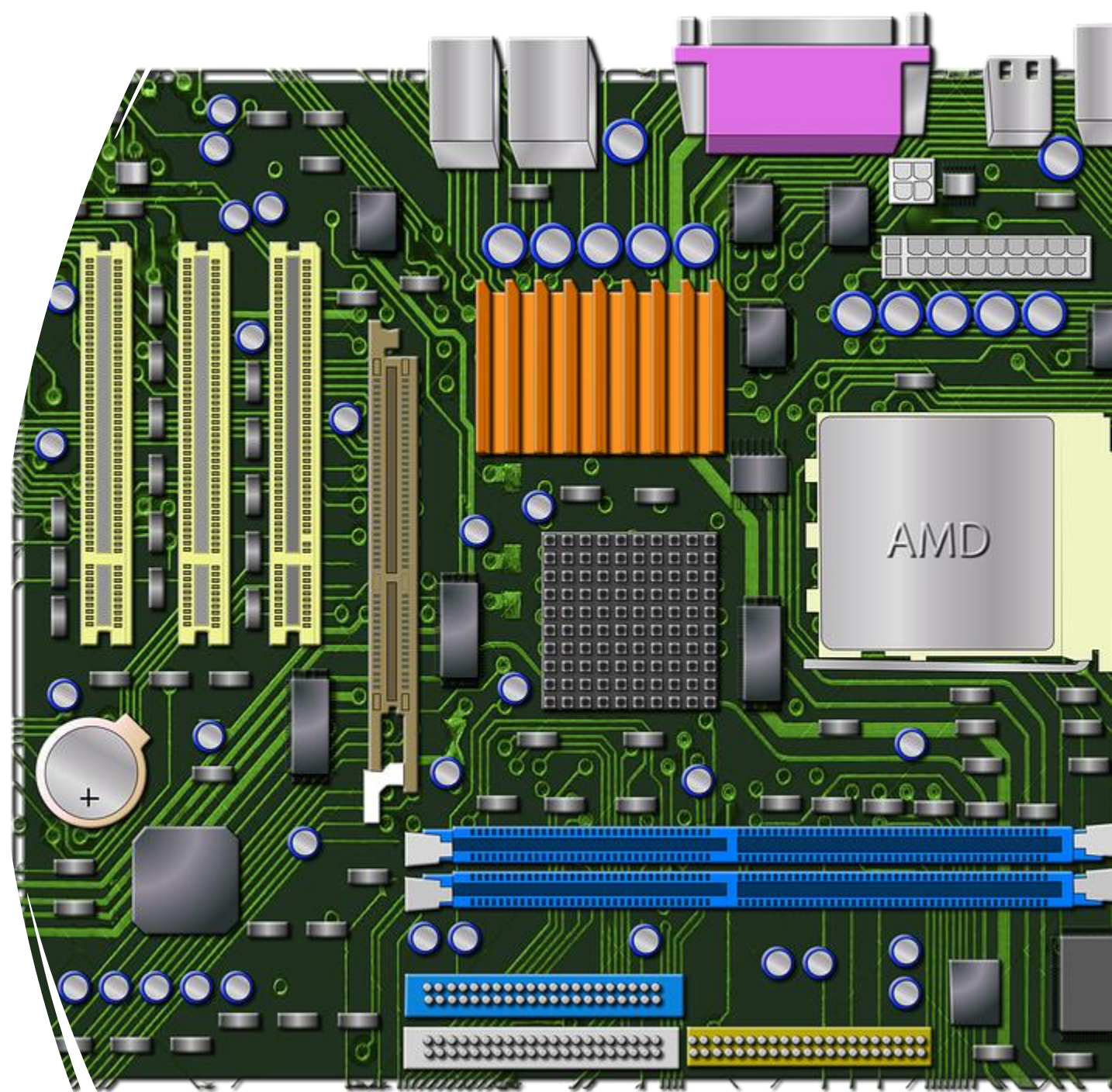


Computer Elements

Physical & non-physical

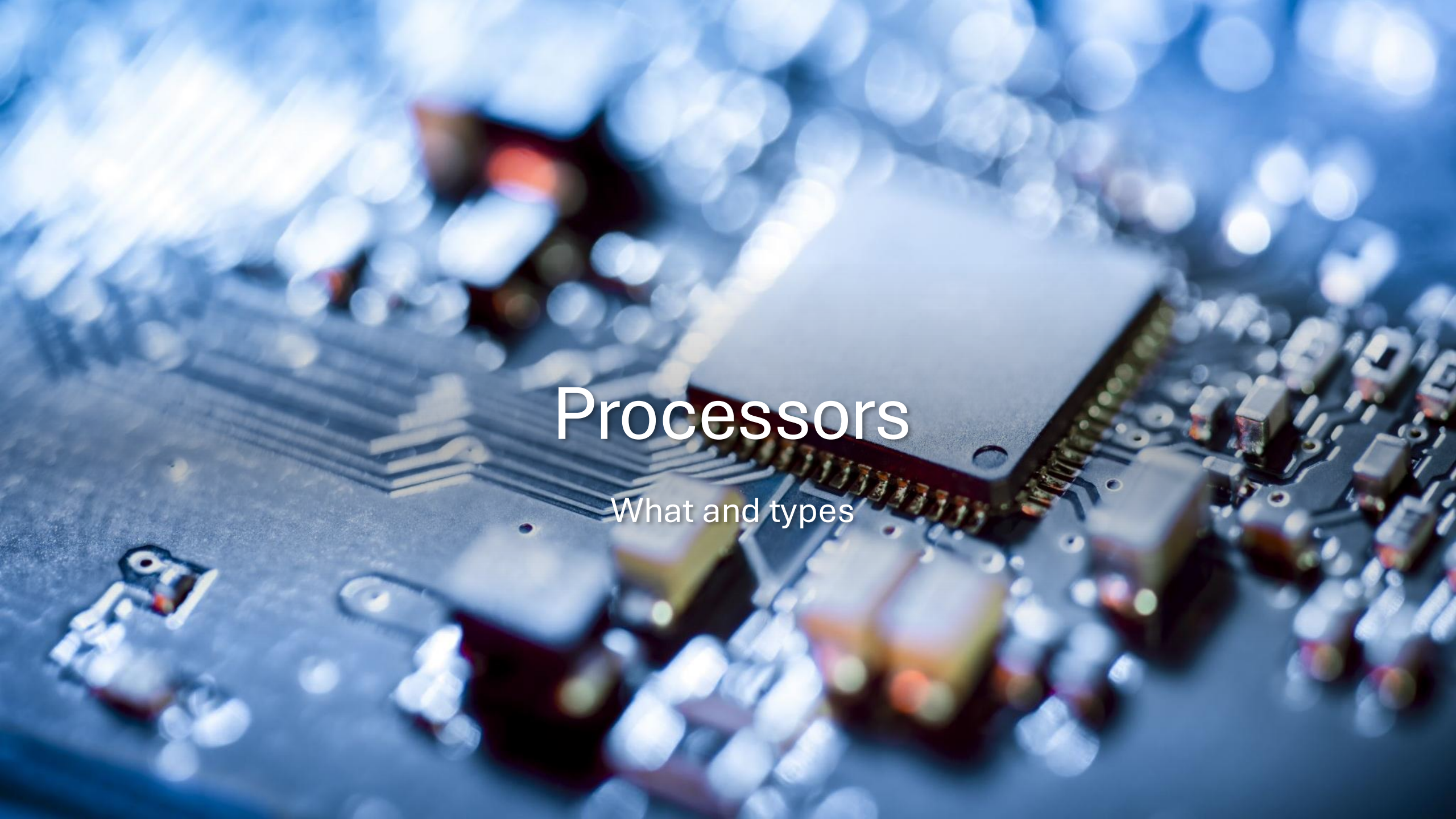
Physical Components

- Mouse
- Keyboard
- Motherboard
- CPU
- Monitor
- Camera
- Speaker
- Microphone
- Hard disk/ROM
- RAM
- Case
- Cooling Fan
- Power Cables
- HDMI/VGA
- Printer
- SSD



Non-physical Components

- Software
- Input Unit
- Output Unit
- GPU
- Arithmetical and Logical Unit



Processors

What and types

Processor

- A processor (CPU) is the logic circuitry that responds to and processes the basic instructions that drive a computer.
- Consists of ALU, FPU, Registers & Cache
 - Single core
 - Dual core
 - Quad core
 - Multi core (upto 12)

Memory

Memory & types

COMPUTER STORAGE OR MEMORY DEVICES



Hard Disk



RAM



ROM



CD/DVD



Floppy



Memory Card



Pen Drive



Tape

Memory

- Remembering / Storing Data or information
- Volatile/Non-Volatile Memory
- RAM/ROM
- DRAM/SRAM



Hardware & Software

Difference and types

Computer Elements

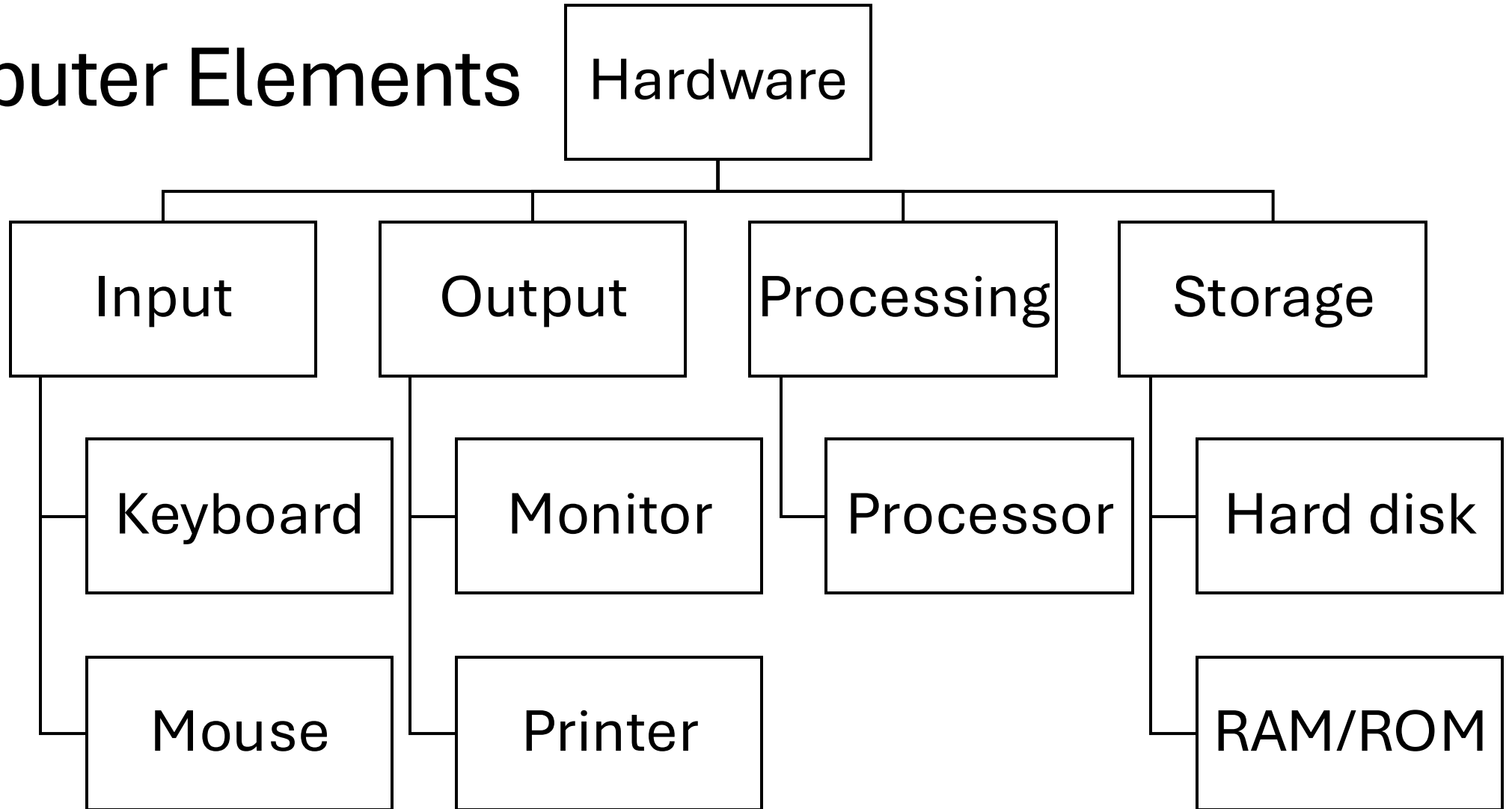
Hardware

- All physical parts of computer
- Can be seen
 - Monitor
 - Mouse
 - Keyboard
 - Joystick
 - Camera

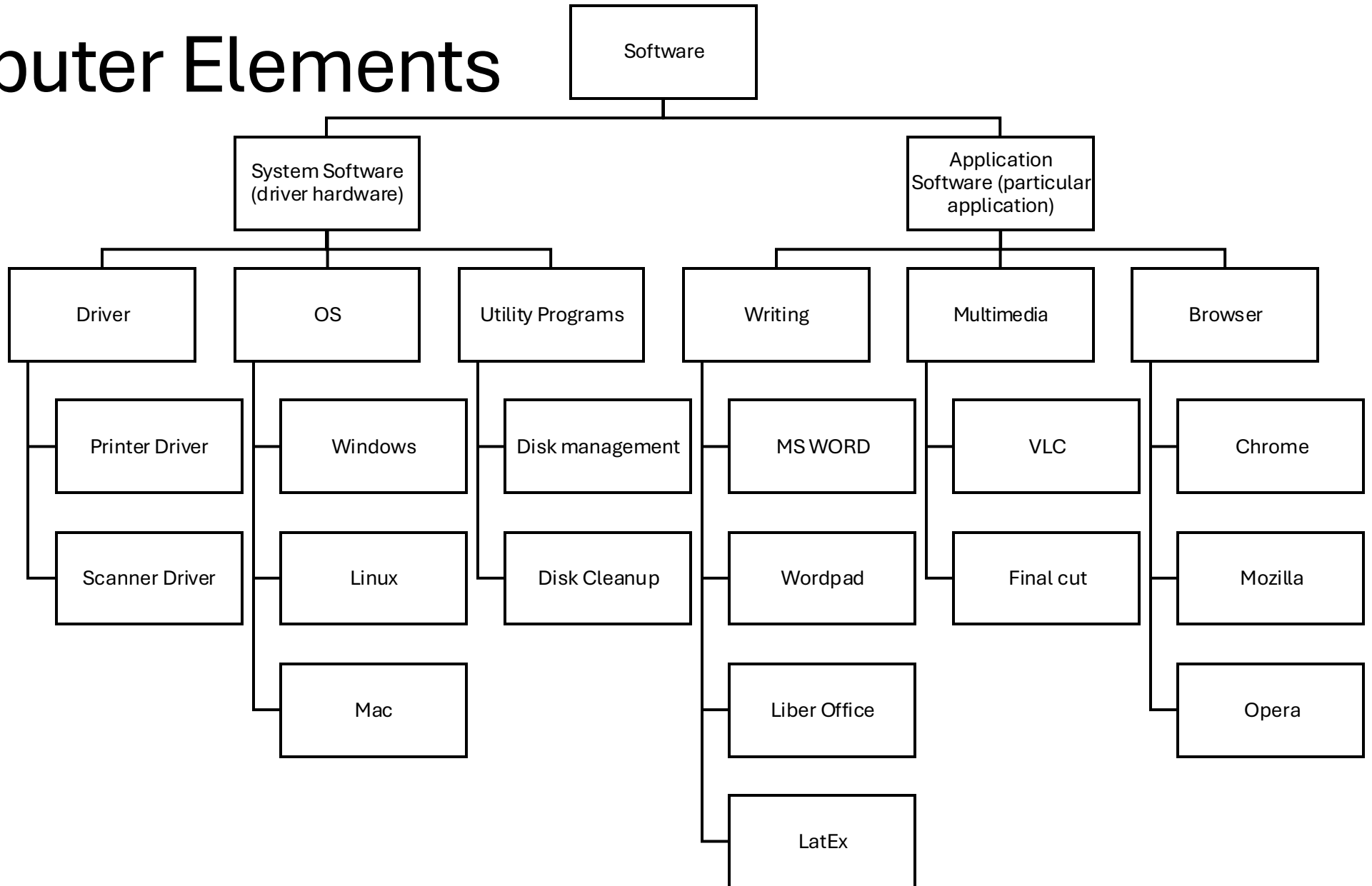
Software

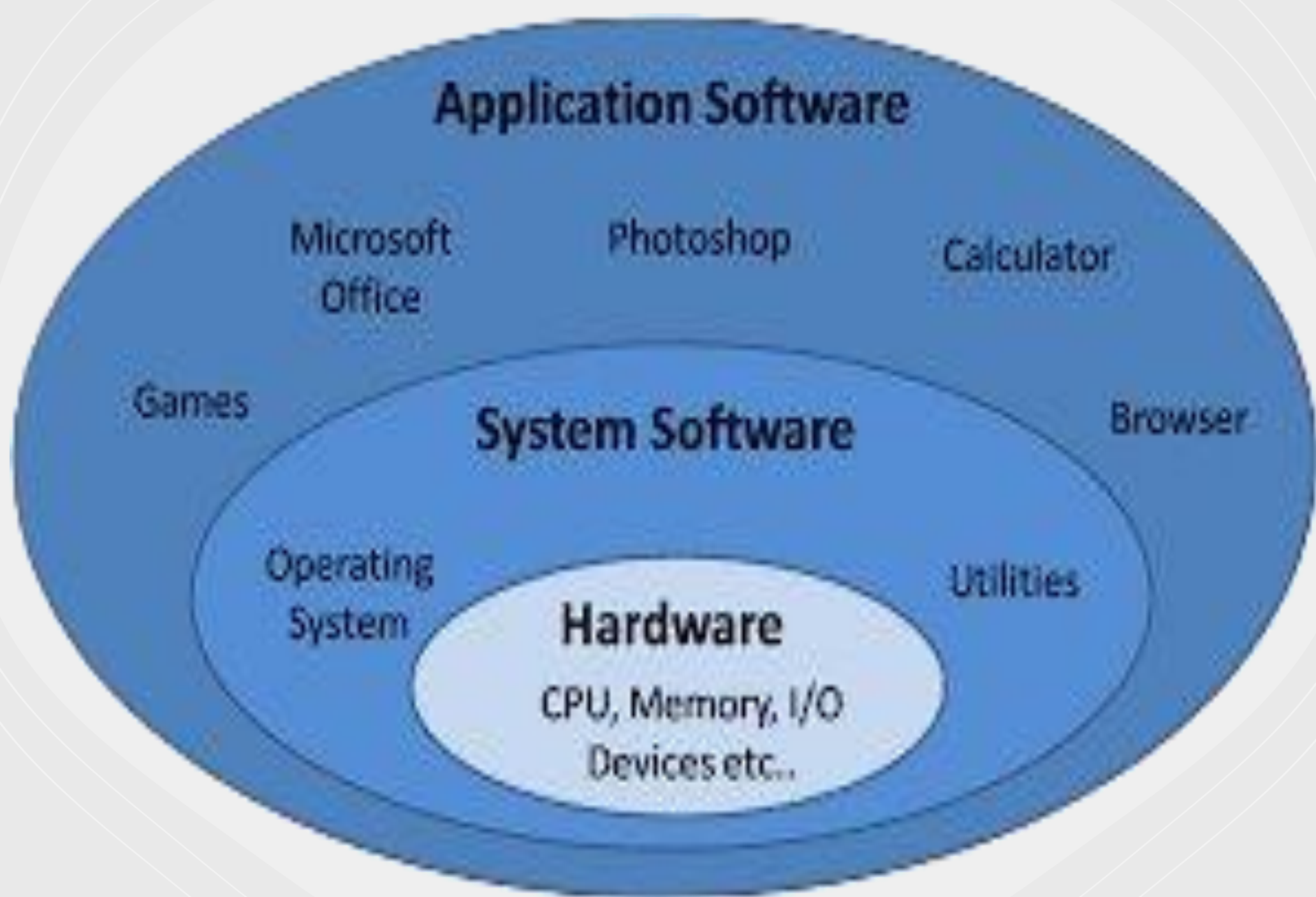
- Instructions or programs that runs the computer
- Can be experienced
 - Browser
 - Anti-Virus Software
 - Multimedia Software
 - WORD/Power Point

Computer Elements



Computer Elements





Types of computer

Based on data processing

Types of Computers

- Minicomputer
- Micro Computer
- Main frame Computer
- Supercomputer

Mini Computers

Originated in 1960s.

- Small mainframes that perform limited tasks.
- Less expensive than mainframe computer.
- Minicomputers are Lower mainframe in the terms of processing capabilities.
- Capable of supporting 10 to 100 users simultaneously.
- In 1970s it contains 8 bit or 12 bit processor.
- Gradually the architecture requirement is grown and 16 and 32 bit.
- Minicomputers are invented which are known as supermini computers.
- e.g. IBM AS400

Microcomputers

- Computers having a microprocessor chip as it CPU.
- Originated in late 1970s.
- First microcomputer was built with 8 bit processor.
- Known as personal computer.
- Designed to use by individual whether in the form of pc's, workstation or notebook computers.
- Small and affordable for general people.
- e.g. IBM PC, IBM PC/XT, IBM PC/AT

Main Frame Computer

- Capable of supporting thousands of user simultaneously.
- Powerful data processing system.
- Can run multiple OSs.
- Process 100 million instructions per second.
- Mainframes are very large & expensive computers having larger internal storage capacity & high processing speed.
- Mainframes are used in the organization that need to process large number of transaction online & required a computer system having massive storage & processing capabilities.

Main Frame Computer

- Mainly used to handle bulk of data & information for processing.
- Mainframe system is housed in a central location with several user terminal connected to it.
- Much bigger in size & needs a large rooms with Air Conditioning System
- IBM & DEC are major vendors of mainframes.
- e.g. MEDHA, SPERRY, IBM, DEC, HP, HCL

Supercomputer

- Most powerful & most expensive computer.
- Used for complex scientific application that requires huge processing power.
- Used multiprocessor technology to perform the calculation very speedy.
- They are special purpose computers that are designed to perform some specific task.

Supercomputer

- The cost of the supercomputer is depended on its processing capabilities & configuration.
- The speed of modern computer is measured in gigaflops, teraflops and petaflops.
 - 1 Gigaflops= 10^9 arithmetic operation per second.
 - 1 Teraflops= 10^{12} arithmetic operation per second.
 - 1 Petaflops= 10^{15} arithmetic operation per second.
- e.g. PARAM , EKA, BLUE GENE/P